MILITARY SPECIFICATION

BUILDING, PREFABRICATED, SECTIONAL, METAL:

TWO STORY, VARIABLE SPAN AND LENGTH

1. SCOPE

1.1 This specification covers two story prefabricated metal buildings of various dimensions with gabled roofs and vertical walls, and a basic floor plan, as specified (see 6.2 and 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

J-C-30	- Cable and Wire, Electrical (Power, Fixed Installation).
L-P-391	- Plastic Sheets, Rods and Tubing, Rigid, Cast, Methacrylate, (Multiapplication).
W-B-30	- Ballast, Fluorescent Lamp.
W-C-375	- Circuit Breaker, Molded Case; Branch-Circuit and Service.
W-C-586	- Conduit Outlets and Entrance Caps, Electrical; Cast Metal - for Shore Use.
W-C-596	- Connector, Plug, Electrical; Connector Receptacle, Electrical.
W-F-406	- Fittings for Cable, Power, Electrical, and Conduit, Metal, Flexible.
W-F-408	- Fitting for Conduit, Metal, Rigid (Thick-Wall and Thin-Wall (EMT) Type).
W-F-412	- Fixtures, Lighting, Incandescent Lamp, Industrial.

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W-F-1234	- Fixture, Lighting (Fluorescent Lamp Industrial).
W-P-115	- Panel, Power Distribution.
FF-H-106	- Hardware, Builders'; Locks and Door Trim.
FF-H-111	- Hardware, Builders'; Shelf and Miscellaneous.
FF-H-116	- Hinges, Hardware, Builders'.
FF-H-121	- Hardware, Builders'; Door-Closing Devices.
FF-W-92	- Washers, Metal, Flat (Plain).
QQ-P-416	- Plating, Cadmium (Electrodeposited).
QQ-S-781	- Strapping, Steel, Flat and Seals.
QQ-Z-325	- Zinc Coating, Electrodeposited, Requirements For.
RR-D-575	- Door, Metal, Sliding and Swinging: Door Frame Metal (Flush and Semi-Flush).
RR-G-661	- Grating, Metal, Bar Type (Floor, Except for Naval Vessels).
RR-W-360	- Wire Fabric, Industrial.
RR-W-365	- Wire Fabric (Insect Screening).
WW-C-581	- Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit: Zinc-Coated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-D-1000	- Drawings, Engineering and Associated Lists.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-F-16081	- Fans, Ventilating, Propeller.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-I-24391	- Insulation, Tape, Electrical, Plastic, Pressure-Sensitive.

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STANDARDS

Federal

FED. STD. No. 141	- Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling, and Testing.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-195	- Marking of Connections for Electric Assemblies.
MIL-STD-810	- Environmental Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN CONCRETE INSTITUTE

ACI 318 - Building Code and Requirements for Reinforced Concrete.

(Application for copies should be addressed to the American Concrete Institute, P.O. Box 4754, Redford Station, Detroit, MI 48219.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANS B18.6.4 - Slotted and Recessed Head Tapping Screws and Metallic Drive Screws.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF CIVIL ENGINEERS

Suggested Specifications for Structures of Aluminum Alloys 6061-T6 and 6062-T6.

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Suggested Specifications for Structures of Aluminum Alloys 6063-T5 and 6063-T6.

(Application for copies should be addressed to the American Society of Civil Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A153	- Zinc-Coating (Hot Dip) on Iron and Steel Hardware.
ASTM A307	- Low-Carbon Steel Externally and Internally Threaded Standard Fasteners.
ASTM A325	- High Strength Bolts for Structural Steel Joints, Including Suitable Nuts and Plain Hardened Washers.
ASTM A449	- Quenched and Tempered Steel Bolts and Studs.
ASTM A490	- Quenched and Tempered Alloy Steel Bolts for Structural Steel Joints.
ASTM B117	- Salt Spray (Fog) Testing.
ASTM D968	- Abrasion Resistance of Coatings of Paint, Varnish, Lacquer and Related Products by the Falling Sand Method.
ASTM D1494	- Diffuse Light Transmission Factor of Reinforced Plastic Panels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF HEATING, REFRIGERATION, AND AIR CONDITIONING ENGINEERS, INC.

ASHRAE Guide and Data Book, Fundamentals and Equipment.

(Application for copies should be addressed to the American Society of Heating, Refrigeration, and Air Conditioning Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code for Unfired Pressure Vessels, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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AMERICAN WELDING SOCIETY

AWS B3.0 - Standard Qualification Procedures.
AWS D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

ARCHITECTURAL ALUMINUM MANUFACTURERS ASSOCIATION

AAMA HSA2 - Specifications for Aluminum Windows.

(Application for copies should be addressed to the Architectural Aluminum Manufacturers Association, 1 East Wacker Drive, Chicago, IL 60601.)

DEPARTMENT OF COMMERCE

CS 214 - Glass Fiber Reinforced Polyester Corrugated Structural Plastic Panels.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ILLUMINATING ENGINEERING SOCIETY

Lighting Handbook.

(Application for copies should be addressed to the Illuminating Engineering Society, 345 East 47th Street, New York, NY 10017.)

METAL BUILDING MANUFACTURERS ASSOCIATION

MBMA - Recommended Design Practices Manual.

(Application for copies should be addressed to the Metal Building Manufacturers Association, 2130 Keith Building, Cleveland, OH 44115.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

STEEL STRUCTURES PAINTING COUNCIL

SSPC - SP3 - No. 3 Power Tool Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburg, PA 15213.)

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STEEL WINDOW INSTITUTE

Recommended Standards for Steel Windows.

(Application for copies should be addressed to the Steel Window Institute c/o Thomas Associates, Inc., Keith Building,Cleveland, OH 44116.)

3. REQUIREMENTS

3.1 Description. The buildings shall be of prefabricated metal, two stories high, with gabled roofs and vertical walls; the size of the buildings shall vary in length and span from 20 feet to 120 feet, as specified. Each building shall be furnished unassembled, and shall include all structural components, outside covering, fasteners, anchor bolts, and materials specified herein except for concrete slabs and foundations. The buildings shall be easily erectable on a prepared concrete foundation by unskilled personnel using handtools.

3.2 First article (first-produced building). The supplier shall furnish one or more buildings as described in 3.2.1 for examination and testing within the time frame specified (see 6.2), to prove that his production methods and choice of design detail will produce buildings that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first-produced examination and tests as specified (see 6.2).

3.2.1 Modified building for the erection test. The modified building for the erection test shall be complete, weatherproof, and similar to each building being furnished. The test building shall consist of three bays, two end bays and one interior bay. The length of the test building shall be not less than 54 feet and not more than 61 feet. The end walls shall have personnel doors located as specified (see 6.2) in both stories and shall be complete with outside stairs, staircases, and stair landings. The end wall and side wall shall have not less than 10 lineal feet of windows located as specified (see 6.2) in each story. The structural steel framing and decking of the second floor, and the interior staircases, stairs, and stair landings shall be installed. The test building shall be erected in accordance with instructions contained in the supplier's erection manual.

3.2.2 Engineering design data and drawings. The following engineering design data and drawings shall be furnished to the Government for approval

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within the time frame specified (see 6.2), prior to the time the first-produced building is furnished (see 6.3.1). Engineering design data including, but not limited to, stress analysis computations and sketches may be on the supplier's format. Drawings shall conform to MIL-D-1000, category E, form I.

(a) Engineering design and structural analysis:

(1) Structural analysis computations and moment and shear diagrams for load carrying members of the structural framework including bracing.
(2) Magnitude and location of reactions for all loading conditions.
(3) Location, magnitude and direction of maximum deflection of primary and secondary structural members due to live load.
(4) Design and details of reinforced concrete footings based on criteria specified in 3.3.2.3.

(b) Drawings:

(1) Details of each connection of primary members, secondary members and primary to secondary members.
(2) Details of the structural framing, length, location and marking of component parts.
(3) Details of first floor slab and footings showing location and placement of reinforcing bars.

3.3 Design criteria.

3.3.1 Dimensional characteristics. Each story of each building shall have a clear ceiling height of 8 feet (see figure 1) and the roof slope shall be 1 in 12. The nominal length of the building shall be expressed in multiples of equal bay lengths and the bay length selected by the supplier shall be within the range of 18 to 25 feet.

3.3.1.1 Interior columns. The interior columns required in the first and second stories or both stories shall be in accordance with the basic floor plan provided by the contracting officer. Typical first floor interior columns are shown in figure 1 for various building widths.

3.3.1.2 Second story structural framing and flooring.

3.3.1.2.1 Structural framing. The structural framing supporting the second floor of the building shall be beam and girder construction and shall be complete with fasteners and connecting devices to attach the flooring.

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3.3.1.2.2 Flooring. The flooring for the building shall be sectionalized and easily fastened to the structural framing. All floors shall be framed and tied into the structural framework to form an integral part of the structure.

3.3.2 Structural design. Except as modified herein, each building shall be designed in accordance with the Metal Building Manufacturers Association's Recommended Design Practices Manual. The supplier has the option of utilizing steel or aluminum for structural framing. Aluminum structural members shall be designed in accordance with the following specifications:

(a) Suggested Specifications for Structures of Aluminum Alloys 6061-T6 and 6062-T6.
(b) Suggested Specifications for Structures of Aluminum Alloys 6063-T5 and 6063-T6.

3.3.2.1 Loads. Unless otherwise specified (see 6.2), each building shall be designed to withstand, in addition to the dead load, the following applied loads:

(a) Roof live load. A roof live load of 40 pounds per square foot acting vertically on the horizontal projection of the roof.
(b) Wind load. Wind velocity (V), V = 70 mph, and q, (impact pressure) in lbs/sq ft $q = .00256V^2$.
(c) Ceiling load. 5 pounds per square foot per story.
(d) Floor load. The second floor structural steel framing shall support 60 pounds per square foot or 100 pounds per square foot as specified.
(e) Stairways. Stairs, stairways, and stair landings shall support 100 pounds per foot.
(f) Additional loads. When required (see 6.2), the contracting officer shall provide the supplier with additional loading criteria for additional static or dynamic loads (see 6.5).

3.3.2.2 Deflections. The maximum live load deflection of primary and secondary structural members, frames, girts, and purlins, and sidewall, endwall and roof panels shall not exceed 1/180 of the span.

3.3.2.3 Foundation design. The supplier shall furnish footing and floor slab designs based on determined reactions for each building (see 6.6). Footings shall be designed for a soil bearing value of not less than 3000 psf, and a bottom-of-footing elevation of not less than 3 feet below the finished grade. Factor of safety against uplift shall be 1.50. The foundation design shall be in accordance with American Concrete Institute Standard 318. The compressive strength of the concrete in 28 days shall be not less than 2500 psi. The footings shall be reinforced with steel and wire fabric.

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3.3.2.4 Floor slabs. Unless otherwise specified (see 6.2), the concrete floor slab of each bulding shall be not less than 4 inches thick reinforced with wire mesh.

3.3.3 Gutter and downspout design. The supplier shall design and furnish a gutter and downspout drainage system. Unless otherwise specified (see 6.2), the gutter and downspouts shall be capable of transporting the flow of water produced by a storm with a rainfall intensity during the time of concentration of 6 inches per hour (see 6.7).

3.3.4 Electrical and lighting system.

3.3.4.1 Electrical system. When specified (see 6.2), the supplier shall design and furnish an electrical system for each building based on criteria of the National Fire Protection Association, National Electrical Code. The supplier shall determine sizing of conductors, types of covering, enclosures and circuit breakers. The system shall be designed to operate at temperatures between minus 20° F. and plus 110° F. without malfunction or damage of any part. The electrical system designed shall be capable of operation utilizing 50 Hz and 60 Hz current. Electrical service required shall be as specified (see 6.8).

3.3.4.2 Interior and exterior lighting. When specified (see 6.2), the supplier shall design and furnish an interior and exterior lighting system based on building use (see 6.9) and shall follow criteria of the Illuminating Engineering Society, Lighting Handbook. The lighting shall be fluorescent, incandescent, or mercury vapor, as specified (see 6.2). Location of lighting fixtures shall be determined by the supplier. Exterior door lighting shall provide each entrance with not less than 5 lumens per square foot at 50 Hz, measured 5 feet from the entrance.

3.3.5 Mechanical design. Unless otherwise specified (see 6.2), each building shall be designed for the attachment of heating, air conditioning, or other environmental equipment. The contracting officer will furnish the size and location of openings for attachment of the equipment to the building (see 6.10). When specified (see 6.2), mechanical equipment shall be furnished by the supplier; the equipment shall be designed in accordance with the American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE) Guide and Data Book, Fundamentals and Equipment.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4.1 Aluminum. All aluminum structural shapes, sheet, and extensions shall conform to applicable American Society for Testing and Materials specifications.

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3.4.2 Structural steel. High strength steel, high strength low-alloy steel, low-carbon steel, and zinc-coated steel sheet shall conform to applicable American Society for Testing and Materials specifications.

3.4.3 Sealant. Sealant, for roof and wall covering side laps, end laps, flashing and trim, shall be pressure sensitive compound. The sealant shall be nonasphaltic, nonshrinking, nondrying, nontoxic and shall adhere to metals and painted surfaces at temperatures from minus 10° F. to plus 140° F. The material shall have a flash point of not less than 300° F. and shall not flow at 200° F. Sealant shall be manufacturers standard, supplied in easily applied paper separated coil stock. Sufficient quantity shall be provided to produce a weathertight seal of all roof and wall joints, trim and flashing, when applied in accordance with the suppliers erection manual.

3.5 Components.

3.5.1 Structural framing. All structural framing shall be accurately prepunched or predrilled for bolted field assembly. When specified (see 6.2), structural members and outside covering shall be prepunched or predrilled to receive sheet metal fasteners.

3.5.2 Outside covering. The outside covering shall be fabricated from zinc-coated steel or aluminum sheets. Aluminum sheets shall be not less than .032 inch thick. The outside covering shall be cleaned, treated and painted with the manufacturers commercial process, color as specified (see 6.2). When tested as specified in 4.5.2.5, the coating system shall not show any evidence of chipping, cracking, peeling, chalking, whitening, or loss of adhesion. The supplier's coating system shall be compatible with those specified in MIL-T-704 and shall be capable of being repainted or touched up without loss of adhesion or lifting of the base coating. Dissimilar metals shall be insulated from each other to prevent galvanic corrosion. Outside covering shall be designed to withstand the loads specified in 3.3.2.1, without buckling, permanent deformation, or separation of joints. Panel joints shall be designed to provide for expansion and contraction caused by temperature differential without causing harmful buckling or opening of joints. Fastening panel to panel, and panel to girts or purlins shall be accomplished with the supplier's standard fastening system, and shall provide a weathertight seal. When tested as specified in 4.5.2.4, there shall be no evidence of leakage, dripping or running flow of water in the interior of the building.

3.5.2.1 Roof covering. Insulated and uninsulated panels shall be the supplier's standard panel configuration. End laps of roof panels shall be not less than 6 inches and shall be fastened together over and to the purlins.

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3.5.2.1.1 Insulated roof panels. When specified (see 6.2), buildings shall be furnished with insulated roof panels. The panels shall consist of a 26 gage steel interior and exterior face, an insulating core, and a joiner member. The "U" factor through the panel shall be not more than 0.10 Btu/hr/ft^2/°F., and the average sound transmission loss through the panel shall be not more than 27 decibels. All connections to structural members shall be made with self-clinching fasteners or other nonexposed fastener systems.

3.5.2.1.2 Uninsulated roof panels. Unless otherwise specified (see 6.2), buildings shall be furnished with uninsulated roof panels. The uninsulated steel roof panels shall be not less than 26 gage. Sidelaps of roof panels shall be not less than one full rib of corrugation. Spacing of panel fasteners at sidelaps shall be not less than 20 inches on center and at endlap shall be not less than 12 inches on centers over purlins.

3.5.2.1.3 Translucent roof panels. When specified (see 6.2), buildings shall be furnished with translucent panels. The panels shall be fabricated from plastic glazing conforming to the Department of Commerce Commercial Standard CS-214, type I, not less than .0625 inch in thickness, and weight not less than 8 ounces per square foot. Color shall be white or green at supplier's option. Configuration, length, and width shall match the roof panel. When tested in accordance with ASTM D1494, the panel shall have a light transmission of not less than 46 percent for white panels and 27 percent for green panels and a smooth finish on both sides. Endlaps shall be not less than 6 inches and sidelaps shall be not less than one full corrugation. The panels shall be compatible with insulated or uninsulated panels.

3.5.2.2 Sidewall and endwall covering.

3.5.2.2.1 Insulated panels. When specified (see 6.2), buildings shall be furnished with insulated sidewall and endwall panels. The panels shall consist of exterior and interior faces of 26-gage galvanized steel sheets. The insulation shall provide a "U" factor of not more than 0.14 Btu/hr/ft^2/°F. and the average sound transmission loss through the panel shall be not more than 27 decibels. All connections to structural members shall be made with self-clinching fasteners or other nonexposed type fastener systems.

3.5.2.2.2 Uninsulated panels. Unless otherwise specified (see 6.2), buildings shall be furnished with the manufacturers uninsulated standard panels. The configuration of corrugation shall be supplier's option. Sidelaps for panels shall be not less than one full rib. Panels should be continuous from base to rake. Where continuity of panels is not in accordance with the design of the supplier's standard building, endlaps shall be not less than 6 inches and shall be made over girts. Spacing of fasteners shall be the supplier's option, and shall provide a weathertight seal.

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3.5.2.2.3 End- and sidelap joints. End- and sidelap joints of wall and roof panels shall be designed to be sealed with sealant to insure water-tightness and to prevent disruption of environmental control within the structure. Insulated wall panels, when applicable, shall be sealed with manufacturers standard sealing devices. Insulated roof panels, when applicable, shall be sealed with a double bead of sealant.

3.5.2.3 Flashing and trim. Each building shall be furnished with the flashing and trim necessary to prevent penetration of dust and water, and to provide a weathertight seal. Flashing and trim shall be preformed and shaped, not less than 26-gage thickness, galvanized steel sheet and shall be provided at the rake, corners, eaves, framed openings and wherever necessary to insure weathertightness and a neat finished appearance.

3.5.3 Closure strips. Closure strips required for weathertightness shall be solid rubber, neoprene, open or closed cell, polyfoam, styrofoam, or closed cell ethylene propylene terpolymer.

3.5.4 Screening. Unless otherwise specified (see 6.2) insect screening shall conform to RR-W-365, type IV, class 1, 18 x 16 mesh. When specified (see 6.2), the insect screening shall be aluminum conforming to RR-W-365, type VII, 18 x 16 mesh. Bird screening shall be of wire fabric conforming to RR-W-360, type I, class 1, 3 x 3 mesh and shall be of low carbon steel wire, galvanized before weaving.

3.5.5 Ventilators. Each building shall be furnished with ventilators complete with bird and insect screens. The color of the ventilators shall match the outside color of the building. The bird screening shall normally be installed toward the building exterior. When tested in accordance with 4.5.2.2.1, the ventilators shall operate easily without binding.

3.5.5.1 Ridge ventilators. Each building shall be provided with continuous ridge ventilators. The ventilators shall be furnished in 10-foot sections, jointed end-to-end, complete with end caps, bird screen, insect screen and a manually-operated adjustable damper. The continuous ridge ventilators shall have a nominal throat opening of 9 or 12 inches, as specified (see 6.2), and shall be operated by a chain and sprocket type mechanism. The adjustable damper shall operate by an adjustable chain loop installed at or near both ends of the building; the chain loop shall be of sufficient length that its lowest point can be adjusted to within 7 feet of the second story floor line.

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3.5.5.2 Louvered ventilators. The buildings shall be furnished with louvered ventilators. The louvered area required and the location of louvers shall be as specified (see 6.2). Louvered ventilators shall be raintight, fixed, overlapping blade type, complete with integral bird screen and insect screen. Frames and blades for louvered ventilators shall be fabricated from 26 gage thickness galvanized steel sheet. The net louver free area of the ventilator shall be not less than 50 percent of the free opening.

3.5.5.3 Sidewall base ventilator. When specified (see 6.2), buildings shall be furnished with raintight sidewall base ventilators, length and location as specified (see 6.2), complete with permanent bird screen and insect screen. Base ventilators shall have a nominal height of 24 inches. The insect screen shall be installed toward the building exterior.

3.5.5.4 Power roof and wall ventilators. When specified (see 6.2), the buildings shall be furnished with power roof and wall ventilators. Power ventilators shall be complete with bird and insect screen, and shall provide 30 changes of total air volume of the building per hour. Roof ventilators shall conform to MIL-F-16081, type I, class 1, style B and shall be not less than 16 inch diameter. Electric motors shall be not more than 1 horsepower and shall be capable of operation on 50/60 Hz, 115 volt service. Motors shall be equipped with overload protection, automatic combination starter and disconnect and remote push button control. Sidewall fans shall be equipped with automatic back draft dampers. The decibel rating of fans shall be not less than the requirements, based on the building use, recommended by ASHRAE, Guide and Data Book.

3.5.6 Doors. Each building shall be furnished with single or double personnel doors, screen doors, horizontal sliding doors, overhead sliding doors or framed openings, as specified (see 6.2). The number, size and location of the doors shall be as specified (see 6.2). When tested as specified in 4.5.2.2.1, doors shall operate easily. All doors, except screen doors, shall be furnished with flashing and weatherstripping to insure weathertightness, and when tested in accordance with 4.5.2.4 shall show no evidence of leakage around the sills, jambs or frames.

3.5.6.1 Personnel doors and frames. All doors shall be hinged to swing outside the building.

3.5.6.1.1 Single personnel doors and frames. Single personnel doors and frames shall conform to RR-D-575, type I, style 1, 1-3/4 inch thick, opening size 3 feet by 7 feet, with continuous honeycomb core.

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3.5.6.1.2 Double personnel doors and frames. Frames and double personnel doors shall be as specified in 3.5.6.1.1. Opening for double doors shall be two openings of 3 feet by 7 feet each. The frame shall be reinforced for door closer mounting.

3.5.6.1.3 Screen doors. Screen doors shall be compatible with the personnel door frames specified herein. Insect screen shall conform to RR-W-365, type IV, class 1, 18 x 16 mesh.

3.5.6.1.4 Glazing. When specified (see 6.2), personnel doors shall be furnished with clear glazing conforming to L-P-391, item A, type I, grade A, thickness 3/16 inch.

3.5.6.2 Horizontal sliding and overhead folding doors. All structural elements, such as door framing, slide track, and other framing shall be fabricated from galvanized steel and completely prefabricated for installation in the specified opening. Structural members of metal doors shall be fabricated from 12-gage galvanized steel sheet and designed to meet all loading requirements.

3.5.6.2.1 Horizontal sliding doors. Each half of double-unit and each single-unit horizontal sliding door shall have at least two self-alining, self-cleaning hangers. Slide track hangers shall be spaced not more than 2 feet on centers. Track hood flashing shall be 26-gage galvanized steel sheet. The doors shall also be guided at the bottom with a self-cleaning door guide. Design and configuration of bottom door guides shall be the supplier's option and shall be compatible with the doors being furnished.

3.5.6.2.2 Overhead sliding doors. Overhead doors shall be of metal or wood, as specified (see 6.2). Tracks shall be galvanized steel, and metal door panels shall be lightweight 20-gage steel box shape. The door shall have a flashing at the bottom to prevent leaks under doors. The doors shall be sealed with a heavy rubber seal at the bottom and the top.

3.5.6.2.3 Pilot doors. When specified (see 6.2), pilot doors shall be placed in horizontal sliding doors and shall conform to 3.5.6.1.1.

3.5.6.3 Door hardware. All hardware for personnel doors shall be given U.S. 10 dull bronze finish in accordance with FF-H-106.

3.5.6.3.1 Hinges. Hinges for personnel and pilot doors shall conform to FF-H-116, type 2115, ball bearing, size 4-1/2 inches by 4-1/2 inches.

3.5.6.3.2 Door closer. Door closer shall conform to FF-H-121, type 3225, style J (modified to be exposed rectangular) size II.

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3.5.6.3.3 Locks. Locks for personnel doors shall conform to FF-H-106, type 810K. Panic bolts for double doors shall be type 822H, and for single doors, type 820H.

3.5.6.3.4 Doorstops. Doorstops shall conform to FF-H-111, type 1320E or 1328E as required.

3.5.6.3.5 Chain bolts. Chain bolts shall be provided on all double personnel doors and shall conform to FF-H-111, type 1021A or 1021B. Bottom strikes shall be provided for the inactive leaf of personnel double doors.

3.5.7 Windows. Unless otherwise specified (see 6.2), the buildings shall be furnished with horizontal slide metal framed windows, number and location as specified (see 6.2). The windows shall be aluminum or steel, as specified (see 6.2), and shall conform to Architectural Aluminum Manufacturers Association HSA2, or the Steel Window Institute, Recommended Specifications for Steel Windows respectively. Metal thickness for principal members of solid section windows shall be not less than 1/8 inch. The metal shall be complete with frames, sash, flashing, hardware, operating devices, fastening devices, screen frames, shutters and other appurtenances necessary for a complete installation and proper operation.

3.5.7.1 Window framing. Jambs, lintels, and sills for windows shall be at the supplier's option and shall support the design load. Jambs, lintels, and sills may be fabricated from aluminum or structural steel; sills shall be contoured to serve as a combination of framing and flashing.

3.5.7.2 Operation of windows. The operable sashes shall be fitted with self-lubricating rollers, providing easy movement of the panel without binding or sticking when tested as specified in 4.5.2.2.1.

3.5.7.3 Window screens and frames. Window screens and frames shall be provided on all operable sash. Window screens shall be readily removable from the exterior of the building. Insect screen shall conform to RR-W-365, type IV, class 1, 18 x 16 mesh.

3.5.7.4 Window shutters. Metal shutters which slide or hinge on both sides shall withstand 70 mph winds. Hardware suitable to close shutters and to hold shutters against the building shall be provided. When tested as specified in 4.5.2.2.1, the shutters shall operate without binding or sticking.

3.5.7.5 Glazing. All windows shall be glazed with 3/16 inch thick, glare reducing, grey tinted, acrylic plastic conforming to L-P-391.

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3.5.8 Fasteners. All threaded fasteners shall conform to the requirements specified herein. The quantity of fasteners furnished shall exceed the actual quantity needed to completely erect the building, as follows:

(a) Nominal size 1/2 inch or larger . . . . . 5 percent extra.
(b) Less than 1/2-inch size . . . . . 10 percent extra.

All fasteners, except anchor bolts, shall be cadmium or zinc-coated steel. Cadmium coating shall conform to QQ-P-416, type II, class 3. Zinc coating shall conform to QQ-Z-325, type II, class 3 for all fasteners 1/2 inch and under. Zinc coating for fasteners larger than 1/2 inch in diameter shall be hot-dip process conforming to ASTM A153.

3.5.8.1 Steel bolts.

3.5.8.1.1 High strength structural bolts. High strength structural bolts shall conform to ASTM A325, A449 or A490, size and strength as required by design criteria.

3.5.8.1.2 Commercial bolts and anchor bolts. Commercial bolts and anchor bolts shall conform to ASTM A307.

3.5.8.2 Plain steel washers. Washers, except those used on the anchor bolts, shall conform to FF-W-92, type A, grade I, class A, cadmium or zinc plated. Washers for anchor bolts shall conform to FF-W-92, type A or B, grade I, class A.

3.5.8.3 Wall and roof fasteners. All sheet metal screws and self-tapping screws shall be cadmium or zinc coated, slotted hexagon head, as shown in ANS B18.6.4, table 50.

3.5.8.3.1 Roof fasteners. Each roof fastener shall be furnished with an attached combination washer. Washers shall be dish type, cadmium or zinc-coated steel, and shall have synthetic rubber, 1/6 inch minimum thickness, bonded to the concave side by a heat- and pressure-type adhesive. The washers shall be attached to sheet metal screws and self-tapping screws with the synthetic rubber face of the washer toward the point of the fastener. The washer shall be of such size that it will fit snugly over the fastener and will not drop off during normal handling by personnel.

3.5.9 Gutters and downspouts. Each building shall be furnished with gutters and downspouts. Gutters shall be furnished with trash screens of 1/2-inch mesh. Downspouts shall conduct water to a height of 8 inches above finished grade and shall be equipped with a 45-degree elbow splash deflector. Spacing of gutter strap supports will be recommended by the supplier and subject to approval of the Government.

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3.5.10 Ceilings. When specified (see 6.2), the building shall be furnished with ceilings. The ceilings shall be furnished with hangers, ceiling panels, and other required components.

3.5.11 Stairs, staircases, and stair landings. Stairs shall be 10-1/2 inch formed steel channel. Stair risers shall be formed steel 7-1/2 inches in height. Anti-slip stair tread nosings that can be anchored into the stair tread shall be furnished. Each staircase and stair landing shall be furnished with a continuous guard rail on each side. The guard rail shall consist of a handrail, an intermediate rail and vertical posts. The handrail shall be not less than 30 inches and not more than 33 inches in height above the nosing of the tread and the stair landing. Intermediate rails shall be placed halfway between the tread nosing and stair landing and the handrail. Handrails shall be returned to walls or posts at the ends of stairways. Handrails, intermediate rails, and posts shall be fabricated from 1-1/2 inch diameter steel pipe.

3.5.11.1 Interior. Each building shall have not less than one interior staircase. Unless otherwise specified (see 6.2), the staircase shall be located in the approximate center of the building parallel to the longitudinal axis. The second floor landing shall be 5 feet by 7 feet and provided with a double guard rail.

3.5.11.2 Exterior. Unless otherwise specified (see 6.2), each building shall be furnished with exterior stairs, staircases, and landings at each end of the building. Stair landings shall be 5 feet by 7 feet with open grating floor, and a double guard rail around the landing and on both sides of the staircase. Open grating landings shall conform to RR-G-661, type optional.

3.6 Electrical and lighting system. The electrical wiring of the electrical and lighting system shall be designed for use inside rigid or flexible metal conduit. Wherever practical, wiring harness shall be provided. The system shall be complete with fasteners and anchorages necessary to attach the conduit and wiring to the structure. When tested as specified in 4.5.2.2.3, the electrical system shall provide the necessary amperage and voltage to the receptacles, wall stitches, luminaires, and electrical equipment. Electric assemblies shall be marked in accordance with MIL-STD-195. When tested as specified in 4.5.2.3, the lighting system shall produce the light specified in 3.3.4.2.

3.6.1 Service entrance. Conductors shall be placed inside zinc-coated conduit conforming to WW-C-581, size as required, with entrance cap conforming to W-C-586, type, design, form, style and size optional.

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3.6.2 Service entrance conductor. Service entrance conductors shall be one of the 75° C. rated rubber or thermoplastic insulated conductors listed in article 310 of the National Electric Code and shall conform to J-C-30, type as applicable, 600 volt single conductor.

3.6.3 Conduit and fittings. Conduit and fittings shall conform to WW-C-581 and W-F-406 or W-F-408, size and number optional.

3.6.4 Power distribution panels. Power distribution panels shall conform to W-P-115, type I, class 2 and shall be equipped with aluminum or copper conductors.

3.6.4.1 Circuit breakers. Circuit breakers shall conform to W-C-375, class 2b or higher interrupting rating.

3.6.5 Conductors.

3.6.5.1 Control wiring. Size of control wiring shall be suitable for the intended purpose and shall not cause malfunction of the control.

3.6.5.2 Wiring conductors. All wiring conductors shall be not less than No. 12 AWG (copper equivalent).

3.6.5.3 Feeder and branch circuit conductors. Conductors for branch circuits shall be 600 volt capacity conforming to J-C-30, type optional, as applicable, plus a grounding conductor. Conductors to ballast shall be in conduit. Feeder conductors which are required to be larger than available conductors of the above types may be interlocked armored cable.

3.6.6 Clamps, straps and fasteners. Clamps and straps holding conduit or cable shall be designed for easy attachment to wood or steel. Fasteners shall also be suitable for securing the clamps and straps to either wood or steel.

3.6.7 Fluorescent fixtures. Fluorescent fixtures shall conform to W-F-1234, style B, kind A, size 1 with mounted ballasts. Ballasts shall conform to W-B-30, 110/125 volt, rapid start, except that the upper limit of operating ambient temperature shall be 110° F.

3.6.8 Mercury vapor fixtures. Mercury vapor fixtures shall be high bay open aluminum for 400 watt or 700 watt lamps. They may have integral ballasts or separate lamp ballasts; power factor shall be not less than 90 percent. Integral ballasts fixtures shall be completely assembled, without lamps.

3.6.9 Incandescent fixtures. Incandescent light fixtures shall conform to W-F-412, type I, size 100 or 200.

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3.6.10 Fluorescent, mercury vapor and incandescent lamps. Unless otherwise specified (see 6.2), fluorescent, mercury vapor and incandescent lamps shall not be furnished with the building.

3.6.11 Receptacles. Receptacles shall conform to W-C-596, style series H. Unless otherwise specified (see 6.2), receptacles shall be placed at 10-foot intervals on all outside walls and not less than 18 inches above the floor. Cover plates shall be surface mounting zinc-coated or cadmium plated steel.

3.6.12 Junction boxes. All junction and outlet boxes shall be 4 inch octagon not less than 1-1/2 inches deep.

3.6.13 Conductor splices. Conductor splices, not on pressure terminals or screwless receptacles, involving size AWG 10 or 12 shall be spliced by insulated twist-on threaded connectors. Splices of AWG 8 and larger shall be spliced by bolt pressure connectors.

3.6.14 Insulating tape. Insulating tape for conductor splices shall conform to MIL-I-24391, 3/4 inch wide.

3.7 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified (see 6.2), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.8 Weatherproof roofing, siding, flashing, doors, and windows. When tested as specified in 4.5.2.4, there shall be no evidence of leaking, dripping, or running flow of water in the interior of the building.

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3.9 Cleaning, treatment, and painting.

3.9.1 Cleaning.

3.9.1.1 Steel. All steel surfaces, except for outside covering shall be cleaned in accordance with Structural Steel Painting Council specification SSPC-SP3, or MIL-T-704.

3.9.1.2 Aluminum. All aluminum surfaces, except for outside covering, shall be cleaned in accordance with MIL-T-704.

3.9.2 Treatment and painting. Structural members shall be treated and painted in accordance with MIL-T-704, type C, color optional.

3.9.3 Erection markings. Each framing component shall be clearly stenciled with erection marking in accordance with MIL-T-704. The marking shall be Gothic capital letters and Arabic numbers not less than 1 inch high. Components too small for stenciling shall be marked with an attached metal tag. Erection markings shall conform to the identification numbers indicated in the erection drawings and the erection manual.

3.10 Instructional material. Three erection manuals shall be furnished with each building. The manuals shall consist of isometric or perspective line drawings and pertinent details showing each successive step in field erection. Included in the manuals shall be a complete bill of material, erection diagrams, a packing list and one set of reduced-size prints of applicable drawings showing the following:

(a) Typical foundation details and anchor bolt layout, including anchorage details.
(b) Wall and roof frame layout.
(c) Wall and roof sheet layout.
(d) Details and typical sections of door, ventilators, base ventilators, stairs, and stair landings.
(e) Layout and details for erection, including, but not limited to, special instructions pertaining to flashing, weathertight trim, ventilation system, and details of the installation of electrical and lighting system, if required.
(f) Instructions for dismantling and re-erection of the building, and components required to accomplish the re-erection of the building.
(g) Concrete floor plans, elevations, sections and details, including all reinforcing details.
(h) General structural notes indicating design loadings and criteria used in design.

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3.11 Identification marking. Each building shall be identified in accordance with MIL-STD-130.

3.12 Workmanship. All parts, components and assemblies of each building shall be fabricated to assure ease of assembly, operation and interchangeability. Components of each building of the same type, class, style or eave height furnished by the supplier under one procurement shall be interchangeable between buildings.

3.12.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.12.2 Welders and welding.

3.12.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

Suppliers who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, supplier's standard welder qualification may be substituted in lieu of the above codes provided that the supplier's procedure is equivalent to the above codes. The supplier shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds. Operators of any automatic, semiautomatic or manual arc-welding machines shall be considered welders for the purpose of this specification.

3.12.2.2 Welding.

3.12.2.2.1 Aluminum. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by inert gas shielded-arc method. Welds may be

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ground, filed, buffed, or chipped but not hammered and shall be free from defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion, or spatter.

3.12.2.2.2 Steel. Surfaces of steel parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when parts connected by welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the building to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 1/16 inch. The welding process shall be at the option of the supplier and shall conform to the provisions of the American Welding Society, D1.1.

3.12.3 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Boltholes in structural members subject to variable loads shall be punched or drilled. Washers shall be provided where necessary.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Acceptability criteria. Buildings which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations and tests in section 4 of this specification will be considered acceptable by the Government.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First-produced building inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 First-produced building inspection.

4.3.1 Sampling. Prefabricated components, accessories, fasteners, anchor bolts, and all material necessary to construct a modified building, (see 3.2.1) shall be selected at random from lots of like items. Supplier's standard commercial items may be furnished when these items meet the requirements of this specification.

4.3.2 Examination. The modified building shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests. The modified building shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples of like components selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. Buildings shall be examined as specified herein for the following defects:

101. Design and fabrication not as specified.
102. Material not as specified.
103. Components missing or not as specified.
104. Treatment and painting not as specified.
105. Stenciling not as specified.
106. Erection manual not as specified.
107. Identification marking not as specified.
108. Workmanship not as specified.

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4.5.2 Tests.

4.5.2.1 Erection. When the supplier conducts the first article inspection, the supplier shall furnish all labor and materials required for the erection of one complete modified building. The modified building shall be made up of randomly selected structural elements and components necessary to erect a complete, weathertight test building. After the modified test building has been erected and tested, the supplier shall completely disassemble the test building and replace all damaged parts and fasteners, retouch damaged protective coatings, and furnish new high strength steel bolts. When all of the above are replaced and repaired, the building from which the components were selected may be accepted for delivery. The outside covering used on the first-produced item test building shall not be furnished as a part of the contract or order. The test building shall be erected in accordance with procedures and instructions described in the erection manual furnished by the supplier. Failure of any component to assemble properly, deformation of components, or inadequate erection manuals shall constitute failure of the test. The supplier shall determine the cause of failures, replace failed components, and assure that all like components are of acceptable quality.

4.5.2.2 Operational tests.

4.5.2.2.1 Doors, windows, ventilators, and shutters. All doors, windows, ventilators, and shutters shall be opened and closed several times to demonstrate that they are easily operable. Difficult operation or inability to operate any doors, windows, ventilators, and shutters shall constitute failure of the test.

4.5.2.2.2 Mechanical equipment. When applicable, all mechanical equipment shall be installed in accordance with the erection manual and cycled not less than five times to prove that the equipment is capable of performing the designed function. Inability to cycle properly or to perform the intended function shall constitute failure of the test.

4.5.2.2.3 Electrical system. When applicable, install the electrical system in accordance with the erection manual and check for continuity of circuits. Inability of the system to provide electrical energy of the required amperage and voltage of receptacles, wall switches, lighting equipment and electrical equipment shall constitute failure of the test.

4.5.2.3 Lighting system. When applicable, install the lighting system in accordance with the erection manual and energize the system. Measure the quantity of light produced by the luminaires at the specified distance above the floor at various locations within the building. Nonconformance to 3.6 shall constitute failure of the test.

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4.5.2.4 Weathertightness. The modified building complete with the specified doors, windows, and appurtenances shall be tested as specified herein. The equipment for testing shall consist of spraying heads capable of spraying water at a uniform rate when suspended 2 feet above the ridge line or held 2 feet away from the walls of the building. The spray heads shall be spaced not more than 2 feet on centers for the full length of the building. The supplier shall provide equipment which will be capable of reducing the pressure inside the building at a rate of 1/4 inch of mercury per 10-minute interval until a total of 1-1/2 inches of mercury reduction from atmospheric pressure has been reached. The equipment shall be capable of controlling and holding the pressure inside the building at 1/4-inch intervals.

4.5.2.4.1 Roof. Place the spraying head arrangement on the roof in such a manner that the spray from the heads allows the water to flow in both directions from the ridge line. Spray water on the building until a sheet flow of water approximately 1/8 inch deep is obtained over the roof surface. After the sheet flow of water develops, reduce the air pressure inside the building at the rate specified in 4.5.2.4. Examine the interior of the building for leaks. Nonconformance to 3.8 shall constitute failure of the test.

4.5.2.4.2 Sidewalls, endwalls, doors, windows, etc. Place the spraying equipment at the eave of the building and spray water directly on the building. When a sheet flow of approximately 1/8 inch is obtained, reduce the pressure inside the building as specified in 4.5.2.4. Examine the sidewalls, doors, windows, and other appurtenances in the sidewall for leakage. Nonconformance to 3.8 shall constitute failure of the test. Repeat the above procedure until all endwalls and sidewalls have been tested.

4.5.2.5 Exterior coating tests.

4.5.2.5.1 Humidity resistance. Humidity resistance of the coating shall be tested in accordance with MIL-STD-810, Method 507, Procedure I. Nonconformance to 3.5.2 shall constitute failure of the test.

4.5.2.5.2 Salt spray resistance. Salt spray resistance of the coating shall be tested in accordance with ASTM B117, using 5 percent salt solution for 500 hours. After exposure of the sample for 500 hours, examine the sample with a magnifying glass for corrosion. A corroded area greater than 5 percent of the total area of the sample shall constitute failure of the test.

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4.5.2.5.3 Impact flexibility. Impact flexibility of the coating shall be tested in accordance with FED. STD. No. 141, Method 6226, except that the testing apparatus shall be a Gardner IG 1120 impact tester or equal. Any evidence of flaking, peeling, or loss of adhesion of the coating after being subjected to 144 inch pounds by the impact tester shall constitute failure of the test.

4.5.2.5.4 Resistance to accelerated weathering. Resistance to accelerated weathering shall be tested in accordance with FED. STD. No. 141, Method 6151, for 500 hours. The gloss of the outside coating shall not change more than 10 percent when tested in accordance with FED. STD. No. 141, Method 6101, 60° gloss test. Gloss shall be tested before and after accelerated weathering. A change of more than 10 percent in the gloss on nonconformance to 3.5.2 shall constitute failure of these tests.

4.5.2.5.5 Coating flexibility. Flexibility of the finish coat shall be tested in accordance with FED. STD. No. 141, Method 6221. Signs of flaking, crocking or loss of adhesion when bent around a 1/8-inch mandrel shall constitute failure of the test.

4.5.2.5.6 Abrasion resistance. Abrasion resistance of the outside covering shall be tested in accordance with FED. STD. No. 141, Method 6191, or ASTM D968, with not less than 20 liters of falling sand. After subjecting the sample to the falling sand, the diameter of the wear-through area shall not exceed 5/32 inch. A wear-through area greater than 5/32 inch shall constitute failure of the test.

4.6 Inspection of preparation for delivery.

4.6.1 Preproduction pack inspection. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 4.0 percent defective.

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109. Materials, methods and containers not as specified. Each incorrect material, method or container shall be one defect.
110. Preservation not as specified.
111. Loose components not packaged as specified.
112. Intermediate packaging not as specified.
113. Packing not as specified.
114. Marking illegible, incomplete or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production packaging and packing that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification. Examination and tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.11).

5.2 Preservation and packaging (see 6.12).

5.2.1 Unprotected surfaces. Threads of anchor bolts and other uncoated steel parts shall be coated with type P-1 preservative in accordance with MIL-P-116.

5.2.2 Small loose components. Fasteners of like size and description such as bolts, nuts, washers, and self-tapping screws shall be packaged in close-fitting fiberboard boxes conforming to PPP-B-636, weather resistant.

5.2.3 Erection manuals. Erection manuals, drawings, and publications shall be preserved in accordance with MIL-P-116, method IC-1. The preserved drawings and publications for each building shall be secured within one of the containers with small loose components (see 5.2.2) in a protected location.

5.2.4 Structural components. Structural components, such as, but not limited to, rigid frames, girts, purlins, and roof and wall covering shall be strapped together in uniform bundles. The grouping of components within the same bundle shall be limited to those of similar shapes, lengths varying by not more than 1/6 the length of the longest component in the bundle, and to a total weight of not more than 3,000 pounds. One end of the bundle shall have the ends of the components flush with each other and as square as components permit. Wood battens shall be used under zinc-coated 1-1/4 inch by .035-inch steel strapping conforming to QQ-S-781, type I. Each bundle shall have a minimum of three straps with the end straps applied 1/6 of the distance from each end of the shortest piece in each bundle and the

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center strap placed equidistant between the end straps. An additional strap shall be required when the distance between the straps exceeds 30 inches.

5.2.5 Intermediate packaging. Building components not bundled in accordance with 5.2.4 shall be intermediate packaged in close-fitting boxes conforming to PPP-B-601, domestic type, style optional. The contents shall be cushioned, blocked, and braced to prevent movement or damage within the boxes.

5.2.6 Electrical components. When furnished, electrical components shall be preserved and packaged in accordance with MIL-E-17555, level A.

5.3 Packing. The complete building shall be packed in a minimum number of crates conforming to MIL-C-3774, type 1, style A.

5.4 Marking. Bundles, boxes and crates shall be marked in accordance with MIL-STD-129. Preservation, packaging and packing marking level shall be level A/A.

6. NOTES

6.1 Intended use. The prefabricated buildings are intended for use such as barracks, warehouses, maintenance shops, post exchanges, laundries and bakeries in the theatre of operations outside the continental United States. These buildings are a part of the Army Facilities Components System (AFCS) and are designed to be erected by unskilled personnel without benefit of special tools or equipment.

6.2 Ordering date. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Basic floor plan required (see 1.2).
(c) Time frame required for submission of the first-produced building and number of units required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier, (see 3.2).
(e) Location of windows and doors in the test building (see 3.2.1).
(f) Time frame required for submission of the engineering design data and drawings (see 3.2.2).
(g) When applied loads are to be other than as specified (see 3.3.2.1).
(h) Additional load criteria when required (see 3.3.2.1).
(i) When the floor slab design shall be other than as specified (see 3.3.2.4).

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(j) When rainfall intensity is other than as specified (see 3.3.3).
(k) When an electrical system design is required and when an electrical system shall be furnished (see 3.3.4.1).
(l) When an exterior and interior lighting system design is required and when a lighting system shall be furnished (see 3.3.4.2).
(m) Whether fluorescent, incandescent or mercury vapor lighting is required (see 3.3.4.2).
(n) When attachments for heating, air conditioning and other equipment is not required and when the supplier will furnish mechanical equipment (see 3.3.5).
(o) When outside covering and structural members are to be prepunched or predrilled (see 3.5.1).
(p) Color of finish coat required for outside covering (see 3.5.2).
(q) When insulated roof, sidewall, and endwall panels are required (see 3.5.2.1.1 and 3.5.2.2.1).
(r) When other than uninsulated roof, sidewall, and endwall panels are required (see 3.5.2.1.2 and 3.5.2.2.2)
(s) When translucent roof panels are required (see 3.5.2.1.3).
(t) When insect screening shall be other than as specified, and when aluminum screening is required (see 3.5.4).
(u) Ventilator throat opening required (see 3.5.5.1).
(v) Area of louvered ventilators required and location of louvers (see 3.5.5.2).
(w) When sidewall base ventilators are required and their length and location (see 3.5.5.3).
(x) When power roof and wall ventilators are required (see 3.5.5.4).
(y) Whether single or double personnel doors, screen doors, horizontal sliding doors, overhead sliding doors, or framed openings are required and their numbers, sizes and locations (see 3.5.6).
(z) When glazing is required (see 3.5.6.1.4).
(aa) Whether metal or wood overhead sliding doors are required (see 3.5.6.2.2).
(bb) When pilot doors are required on horizontal sliding doors (see 3.5.6.2.3).
(cc) When other than horizontal slide metal framed windows are required (see 3.5.7).
(dd) Number and location of windows required (see 3.5.7).
(ee) Whether aluminum or steel windows are required (see 3.5.7).
(ff) When overhead ceilings are required (see 3.5.10).
(gg) When the staircase shall be located other than as specified (see 3.5.11.1).
(hh) When the exterior stairs shall not be furnished as specified (see 3.5.11.2).

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(ii) When fluorescent, mercury vapor and incandescent lamps shall be furnished (see 3.6.10).
(jj) When the receptacles shall be placed other than as specified (see 3.6.11).
(kk) Composition required (see 3.7). Composition II should be required when the application of varnish is accomplished within a geographical area required to comply with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.
(ll) Time frame required for the submission of the preproduction pack (see 5.1).

6.3 First-produced building. Any changes or deviations of production buildings from the approved first-produced building during production will be subject to approval of the contracting officer. Approval of the first-produced building will not relieve the supplier of his obligation to furnish buildings conforming to this specification.

6.3.1 Engineering design data and drawings. The contracting officer should forward to the engineering support activity two copies of the engineering and design data and two complete sets of reproducible drawings.

6.4 Basic floor plan. The contracting officer should provide the supplier with a basic floor plan of the building when the building configuration is known.

6.5 Additional loads. The contracting officer should furnish the supplier with design requirements for additional dynamic and static loads when requested by the procuring activity. The additional loads could be in the form of air conditioning equipment or other machinery which attaches to the building.

6.6 Foundation. The supplier shall furnish only the design for the footings and floor slabs; the supplier is not required to furnish any material for the footings or floor slabs.

6.7 Gutter and downspout design capacity. The quantity of water to be transported from the roof of the buildings shall be determined utilizing the rational method of analysis as shown in the following formula:

Q = Air

Q = Quantity of water in CFS
A = Area in acres
R = Constant for the type of surface
i = Rainfall intensity during time of concentration in inches per hour

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The rainfall of 6 inches per hour should not be changed unless definite information is available regarding the building location. The contracting officer should refer questions regarding rainfall intensities to the Commanding General, U. S. Army Mobility Equipment Command, Directorate of Research, Development and Engineering, Fort Belvoir, VA 22060, ATTN: AMSME-RZM-MKB

6.8 Electrical service. The contracting officer should furnish the supplier with requirements for electrical service to the buildings. The information should include the total voltage and amperage service required and the number of electrical outlets required for 110/120 and 220/240 volts service (see 3.3.4.1).

6.9 Internal illumination. The contracting officer should furnish the supplier with illumination requirements based on building use. The information should provide the supplier with the number of lumens per square foot required and the distance above the floor this illumination is required (see 3.3.4.2).

6.10 Mechanical equipment for buildings. The contracting officer should provide the supplier with the required openings and connections for attachment of military designed heating or air conditioning equipment. When mechanical equipment is to be designed by the supplier, the contracting officer should furnish the supplier with the engineering data and guidance (see 3.3.5).

6.11 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation of preserve, package, pack, and mark the prefabricated buildings in accordance with this specification.

6.12 Preservation, packaging, and packing. No levels of preservation, packaging, and packing are specified, as the buildings are intended to be used in field operations, are destined for outside CONUS use, and are not intended for immediate use in CONUS operations. However, the buildings may be held in indefinite storage thereby requiring level A preservation, packaging, and packing.

Custodian:

Army - ME

User activity:

Army - CE

Preparing activity:

Army - ME

Project No. 5410-A128

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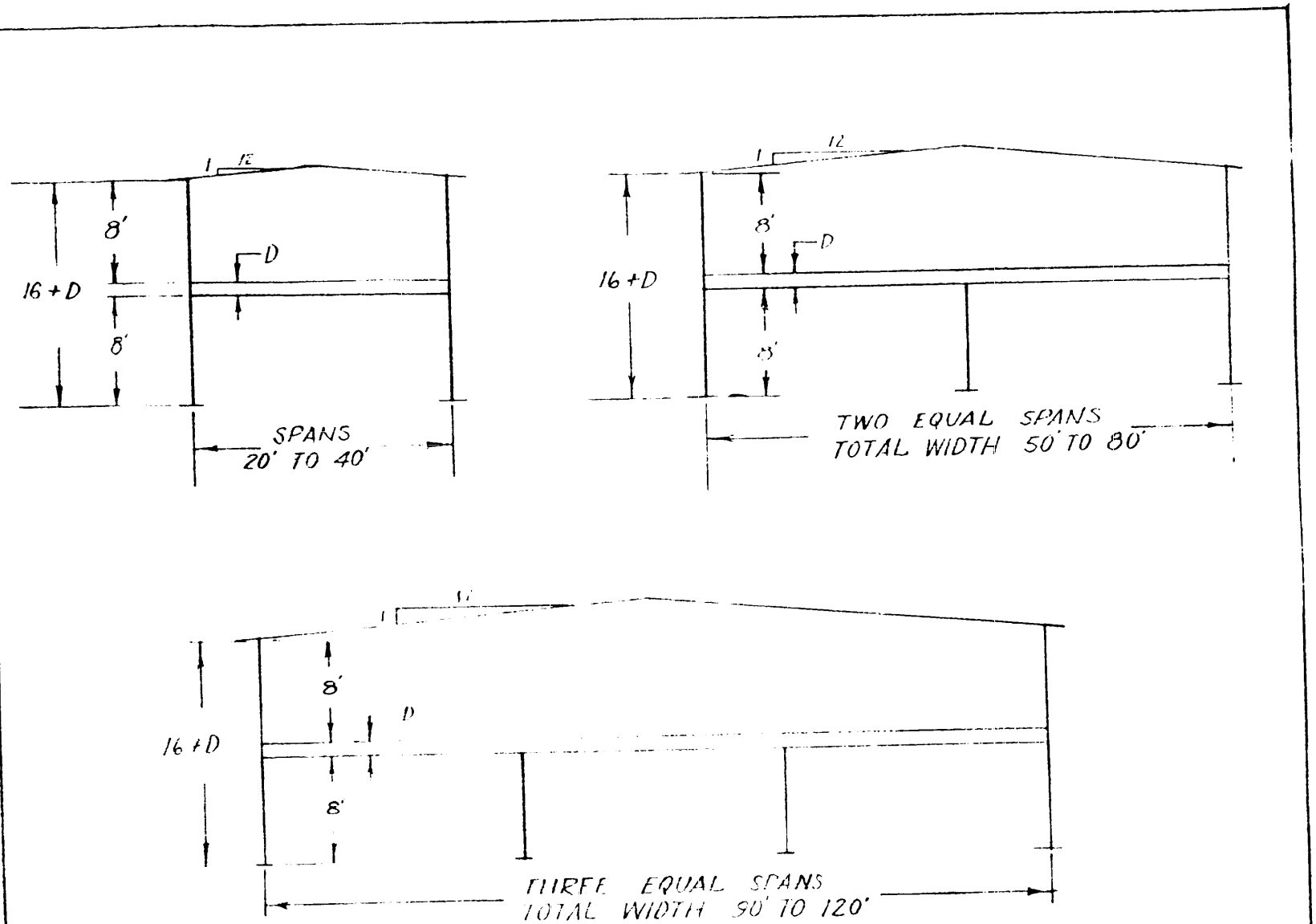


FIGURE 1 BUILDING CONFIGURATIONS, TYPICAL

X-2237

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-B-52755(ME) Building, Prefabricated, Sectional, Metal: Two Story, Variable Span and Length

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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